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Libet's intention reports are invalid: A replication of Dominik et al. (2017)

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A R T I C L E I N F O

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A B S T R A C T

In the "Libet experiment" the onset of movement-related brain activity preceded the reported time of the conscious intention to move, suggesting that conscious intention may not play a role in initiating voluntary movements (Libet, Gleason, Wright, & Pearl, 1983). Dominik et al. (2017) provided evidence that the intention reports employed in the Libet experiment, which Libet et al. (1983) found to precede movement reports, are invalid. In the study by Dominik et al., intention reports preceded movement reports only when participants had prior experience making movement reports. Individuals without such experience reported intention around the same time as movement. These findings suggest that Libet's intention reports do not reflect experiences of intention, but, rather, inferences based on prior experience with movement reports. Our study replicated the core findings of Dominik et al. We argue that Libet's intention reports are invalid and explore the phenomenology of intention in the Libet experiment.

1. Introduction

1.1. The Libet experiment

The "Libet experiment" examined the temporal relationship between two events associated with voluntary movement: the conscious intention to move and electroencephalographic (EEG) activity believed to reflect cerebral processes preparing the movement (Libet et al., 1983).[2] Libet et al. (1983) provided evidence that the "readiness potential" – an EEG signal believed to precede voluntary movements and reflect movement-related brain activation (Kornhuber & Deecke, 1965; Gilden, Vaughan, & Costa, 1966) – began about 350 ms earlier in time than the conscious intention to move. The implication was that conscious intention could not play a causal role in movement initiation, since movement preparation significantly preceded the time of conscious intention.

Banks and Pockett (2007, p. 659) noted that most replications of the Libet experiment have found the same temporal relationship between the readiness potential (RP) and conscious intention (e.g., Keller & Heckhausen, 1990; Trevena & Miller, 2002). In addition, functional magnetic resonance imaging (fMRI) variants of the experiment have reproduced this temporal relationship (e.g., Lau, Rogers, Haggard, & Passingham, 2004). However, many researchers have criticized and/or brought evidence against various aspects of the experiment, including its assumptions (e.g., Nachev & Hacker, 2014), conceptual basis (e.g., Papanicolaou, 2017), empirical

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[2] For the sake of simplicity, we refer to the components of voluntary movements (e.g., intentions, cerebral activity) as "events" unless otherwise noted.

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basis (e.g., Schurger, Sitt, & Dehaene, 2012), methodology (e.g., Verbaarschot, Farquhar, & Haselager, 2015), and ecological validity (e.g., Navon, 2014). As a result, serious doubts have arisen about the Libet experiment's basic results and typical interpretations.

A fundamental issue is that the Libet experiment did not measure "intention" directly.[3] Intention was measured indirectly, via participant reports of intention. Libet et al. (1983) devised a method to allow reports of the conscious intention to move, thereby operationalizing it: participants reported their intentions by tracking time on a clock and later reporting the moment of intention. This provided a means of comparing a subjective conscious intention to the "objective" time that brain activity began (i.e., the RP, as measured via EEG). A crucial question remains, however: do the reports yielded by this indirect method truly and accurately reflect intentions? Dominik et al. (2017) provided evidence suggesting that these reports are invalid measures of intention.

1.2. Methodology of the Libet experiment

The relevance of the findings of Dominik et al. (2017) necessitate a more detailed understanding of the Libet experiment. Libet et al. (1983) trained six individuals to complete three distinct tasks. As participants completed tasks they monitored a revolving dot on an oscilloscope set up as an analog clock. On each task participants noted, and then later reported, the location of the dot at the time when they estimated a certain event to have occurred. The "S" report was of the moment a tactile *stimulus* was delivered on the back of the hand. The "W" report was of the moment one *wanted* or "felt the urge" (i.e., intention) to make a spontaneous wrist movement (Libet et al., 1983, p. 625). The "M" report was of the time one made a spontaneous wrist *movement*. The timing of W and M reports was measured in reference to the objective time of movement, which was determined by electromyographic (EMG) recording. Participants were given strict instructions and training on all experimental tasks.

Participants' S reports were used to "correct" their W and M reports for any bias associated with the clock method of reporting (Libet et al., 1983, p. 631).[4] Since the timing of the delivered stimulus was objectively measured, S reports could be compared to this measure in order to calculate their accuracy. This bias averaged to about −50 ms across participants (i.e., participants reported S times as 50 ms earlier than when the stimulus was actually delivered). A corresponding correction was applied to grand average W and M reports, resulting in an average W report of about −150 ms (i.e., 150 ms prior to movement) and an average M report of about −40 ms (i.e., 40 ms prior to movement). Across participants, W reports consistently preceded M reports, and this difference averaged to about 115 ms. The RP, beginning around −500 ms, preceded W reports by about 350 ms. As noted, these results have been widely replicated in both EEG- and fMRI-based experiments. Those variants of the experiment which allow participants some additional choice in addition to the "choice" of when to move spontaneously, such as whether to move the right or left hand, have also produced similar results (e.g., Haggard & Eimer, 1999).

1.3. The study by Dominik et al. (2017)

Dominik et al. (2017) developed a variation of the Libet experiment which focused solely on W (intention) and M (movement) reports in order to test the validity of W reports. These authors sought to determine whether W reports are influenced by prior experience with M reports and knowledge of the distinction between M and W reports. In this study, 20 participants were assigned to one of two groups and asked to complete two tasks. One task required M reports and the other required W reports. Trials for the two tasks were not interspersed but separated into independent blocks. The groups differed only in the order in which they completed tasks. The movement-first (M-First) group completed the M task prior to the W task, while the intention-first (W-First) [5]group completed the W task prior to the M task. Crucially, when completing the first task, participants in each group were given minimal instructions and were not informed of the nature of the second task.

Results showed that means did not differ between the groups on the first task completed, despite the fact that one group reported intentions and one group reported movements. The means relative to the objective time of movement for each group were around −10 ms on the first task, so both groups made reports nearly identical in time with the actual movement. On the second task completed, there was a significant within-groups difference in reports in the M-First group but not the W-First group: those in the M-First group reported W significantly earlier in time than M, but there was no difference in W and M reports in the W-First group.

Dominik et al. (2017) argued that their findings were evidence that Libet's intention reports are based on inference rather than an experience of intention. As part of this general conclusion, they argued that there is only one "event" to report in the Libet experiment: movement. Dominik and colleagues noted that participants in the Libet experiment participants were trained on, and gained experience with, both M and W reports prior to data collection. In their own study, however, participants who made intention reports first (i.e., in the W-First group) did so without the influence of such training and experience. As a result, these "intention" reports were made near the time of movement and not significantly different from reports of movement. Pointing to results on the second task, Dominik et al. (2017) speculated that once individuals gain experience making movement reports, they *infer* that their intention reports should precede these in time. Instead of actually experiencing intention, these individuals may, on some level, simply come to believe that, logically, intention should occur before movement. Consequently, they report W earlier than M and only

[3] It is unclear whether it is logically or nomologically possible to "measure intention directly."

[4] Libet et al. (1983) controversially assumed that any common biases associated with the revolving dot method of reporting would be stable across W, M, and S reports.

[5] M-First and W-First are terms we have introduced, signifying which task was completed first. They did not appear in the paper by Dominik et al. (2017). We employ them in this paper to make our references to groups more intuitive.

report M near the objective time of movement. Dominik et al. (2017, p. 260) called this process "anchoring" and contended that in the Libet experiment W reports are an "artificially induced impression" produced by "mere experience with M measurements." The clearly experienced phenomenon of movement, and reports thereof, may serve as an objective reference point; if individuals do not actually experience intention, they may infer that it must have occurred *prior* to when they moved. Participants gain experience making M reports, which they assume to coincide with the actual time of movement, and then use these reports as an anchor point from which they can place W reports earlier in time.

To further bolster their argument, Dominik et al. pointed out that an anchoring effect occurred only in the M-First group. In other words, those who first made M reports had their experience with these reports to anchor their W reports (i.e., they could report W earlier in time than M).[6] Those in the W-First group, however, had no experience with M reports when making W reports and, consequently, made indistinguishable W and M reports, both of which were near the time of movement. It is worth noting, as discussed by Dominik et al. (2017, pp. 256-257), that the Libet experiment itself showed a similar order effect despite extensive training of participants. On average, W reports made after M reports were significantly earlier in time (about 50 ms) than W reports made prior to M reports (Libet et al., 1983, p. 632).

Dominik et al. (2017) contended that their findings were provisional evidence that intention reports in the Libet experiment lack validity, since they appear to be inferences rather than genuine reports. On their view, though the Libet experiment appeared to show "a difference between two introspective phenomena," this difference was actually between one such phenomenon (i.e., the experience of movement) and an artificial impression of "intention" (Dominik et al., 2017, p. 260). Other studies within the Libet paradigm have suggested that intention is "inferred" (e.g., Pockett & Purdy, 2011; Rigoni, Brass, & Sartori, 2010; also see Eagleman, 2004). Banks and Isham (2009), for example, provided evidence that intention reports in the Libet experiment are retrospectively inferred based on the time of movement and can be influenced by pairing the movement with a delayed auditory tone. Reports were shown to be more delayed as the tone was more delayed from movement time. Although these authors concluded that their results were evidence that conscious intention does not initiate movement, the results might also lead to the conclusion that obtained intention reports are an invalid artifact of the experimental setup (i.e., intention is being inferred and, thus, "intention" is not actually being measured).

Note that it is possible to accept the general conclusion of Dominik et al. (2017) – that Libet's intention reports reflect invalid inferences – without accepting their full explanatory account of the inferential process. Their account (outlined above) includes the claim that movement is the only genuine introspective impression to report, which entails that intention is phenomenally absent from the spontaneous voluntary movements required in the tasks. This is only one possible interpretation of the anchoring effect, however, and does not appear essential to it. An alternative interpretation is that an experience of intention *is* present in these movements, and that only the alternate experimental design pioneered by Dominik and colleagues has accurately captured reports thereof. That is, perhaps participants naturally experience intention at more or less at the same time as movement, and intention reports in the W-First group genuinely reflect this experience, while in the M-First group the anchoring effect overrides or taints this experience and/or the ability to report it. It may even be that intention is experienced but generally unamenable to accurate measurement via indirect reports.

1.4. The current study

The findings and conclusions of the Libet experiment and its replications center on the temporal relationship of the RP and intention. The validity of this relationship depends on the validity of both the RP and intention reports. The basic question raised by Dominik et al. (2017) is whether the intention reports pioneered by Libet et al. (1983) are valid measures of intention. If the manner in which the Libet experiment measured intention is invalid, then the temporal relationship between the RP and intention is invalidated. Given the breadth of empirical and theoretical research based on the Libet experiment, as well as its implications for various scientific and philosophical discussions, such as those involving free will, the findings of Dominik et al. warrant serious consideration. The current study is an attempt to replicate these findings and explore their implications.

We closely mirrored the aforementioned methodology of Dominik et al. (2017), but we made some additions. First, we included a stimulus (S) task similar to that employed by Libet et al. (1983), which served two purposes. It measured participants' average error in using the analog clock to make reports, and it was a "dummy task" to allow experience with the clock-based reporting method prior to any M or W trials, per the recommendation of Dominik et al. (2017, p. 262). Second, we collected EMG data to measure the precise moment of movement onset. Specifically, we used the EMG signal to determine when participants' finger movement began during a mouse click. The mouse click initiates a digital timestamp in the software programs, but this timestamp is inevitably delayed in time past the earliest moment of movement, since completing the press of a mouse button necessarily takes more time to execute than simply initiating the first movement to press the button. Thus, we used EMG data to correct our digitally recorded timestamps. However, since Dominik et al. did not employ EMG corrections, we analyzed our data both with and without EMG corrections and compared results.

We expected to corroborate the results of Dominik et al. (2017). Specifically, we hypothesized that: (a) when participants in each group are naïve about the distinction between moving and intending to move and have no experience making reports thereof, intention reports and movement reports would not differ between groups, (b) intention would be reported earlier in time than movement only in the M-First group (i.e., only when participants report intention after having reported movement), and (c) intention would be reported earlier in the M-First group than in the W-First group.

[6] Note that the methodology and timing of W reports in the M-First group mirrors what normally occurs in the Libet experiment for all participants. Participants make W reports after becoming familiar with M reports, and, on average, W is reported significantly earlier in time than M.

2. Materials and methods

2.1. Participants

We collected data from 22 students (6 males; 3 left-handed; aged 18–64 years, $M_{age} = 22.55$) from Eastern Kentucky University. We collected data from an additional 7 students which were excluded due to: (a) unanalyzable EMG data (n = 4), (b) researcher error (n = 2), and (c) having heard of the Libet experiment (n = 1).[7] All students received three outside activity credits for 1.5 h of participation. Before beginning the study, each participant was required to affirm that she/he possessed normal or corrected vision and hearing, did not have a learning or reading disability, and did not have or a current or prior neurological or psychological disorder.

2.2. Forms and technical equipment

All participants completed a consent form and a demographic form. The information from the latter was used to understand the nature of the data sample and included gender, age, handedness, race, and whether a participant had taken any medication or substance that might impact ability to focus.

We presented all visual stimuli (white on gray background) with E-Prime 2.0 software (Psychology Software Tools, USA). Following the example of recent research (e.g., Jo, Hinterberger, Wittmann, Borghardt, & Schmidt, 2013), a Libet-style analog clock with a diameter spanning a visual angle of approximately 3° was presented on a computer screen.[8] The visual angle of 3° was approximated by standardizing the setup of equipment (i.e., maintaining the monitor, desk, and chair in the same position for each participant) and having participants' eyes approximately 67 cm away from a 3.5 cm clock face. The clock hand rotated clockwise with a revolution period of 2.55 s, which closely mirrored that of Dominik et al. (2017), 2.56 s. A new clock "tick" occurred every ~16.667 ms.

We recorded behavioral data with E-Prime 2.0 software (Psychology Software Tools, Inc.) and physiological data with AcqKnoweldge 3.9.1 software (Biopac Systems, Inc.). These software programs ran on two linked HP ProDesk 600 G2 MT computers (HP Inc.). Participants viewed a BENQ model RL2455 computer monitor with a 60 Hz refresh rate and resolution of 1280 × 1024. We presented auditory stimuli with Dell A225 speakers, standardized at 100% system volume and 50% speaker volume. Participants used a Logitech M100 USB mouse on all tasks. Tests revealed that the mouse had a maximum polling rate of approximately 125 Hz (i.e., maximum temporal resolution of 8 ms) on the computer.

We collected and analyzed electromyographic (EMG) data via a Biopac MP100A recording system (Biopac Systems, Inc.) employing EEG100C amplifiers with a gain of 20,000, a 35 Hz LPN (low pass notch) filter, and 0.1 Hz HP (high pass) filter. We used two Biopac EL258 Ag-AgCl (silver-silver chloride) unshielded electrodes with holes. Following standard procedure, satisfactory electrode impedance level was 5 KΩ or below.

2.3. Procedure and design

2.3.1. General setup

Each participant completed the study independently. Upon arrival, participants provided an identification code. After reading the consent form and discussing it verbally with the researchers, participants provided informed consent. They then provided demographic information. Next, researchers connected participants to the EMG equipment. All tasks required moving and pressing the left button of a computer mouse. Though handedness was determined prior to task completion, we required all participants to use a right-handed mouse, so all EMG connections were to the right hand and right forearm (Fig. 1).

Participants completed three distinct tasks. Prior to arrival participants had been randomly assigned to one of two groups, which determined the order of task completion (Fig. 2). All tasks employed the analog clock and consisted of training trials, regular trials, and a one minute break halfway through. On each trial the clock face appeared for a randomly assigned length of time (1000–2000 ms) before the clock hand appeared and started rotating. The clock hand could begin rotating from any location. Participants were to monitor the clock hand and note its location when a certain event occurred. After the event, participants made a mouse click in order to stop the clock hand's rotation. After this click, the rotation continued for a randomly assigned length of time (1000–2000 ms). On the next screen, only the clock face appeared. Participants reported the clock hand location they had previously noted; they clicked on the clock face to indicate where the clock hand was when the event occurred. The event depended on which task was being completed (see task descriptions below). This method of recall and reporting, in which participants simply indicate where the clock hand was at the moment an event occurred, was termed the "absolute mode" by Libet et al. (1983, p. 626). We followed the example of Dominik et al. (2017) and used only this mode of recall.[9]

Researchers gave participants general instructions for using the clock to make reports before they completed trials. Participants were to note the clock hand's location while keeping their eyes fixated on the center of the clock rather than following the clock hand

[7] After the study concluded, researchers asked participants whether they were familiar with the Libet experiment.

[8] The clock was virtually identical to that used by Jo et al. (2013). These researchers were generous enough share their coded E-Prime file, which included a programmed clock.

[9] Libet et al. (1983, p. 626) compared this mode with an "order mode," in which participants reported whether the final location of the dot, which continued to move after the event occurred, was earlier or later on the oscilloscope than the location of the dot when the event in question (W, M, or S) occurred. Libet et al. found no meaningful differences in results between the absolute and order mode.

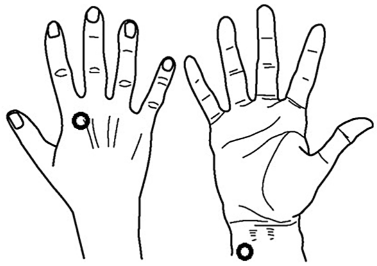

Fig. 1. Placement of EMG electrodes (image modified from Lameira et al., 2009). Pilot testing prior to data collection revealed that optimal EMG data could be obtained using one electrode behind the trapezoid bone (i.e., lower knuckle) of the right index finger and one electrode of the ventral right forearm just below the wrist.

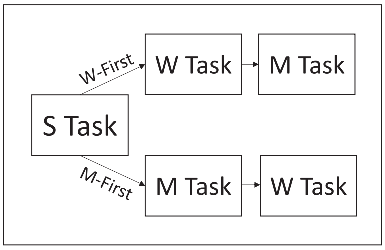

Fig. 2. Task order by group. All participants first completed the S Task (report of auditory stimulus), but participants differed in the order in which they completed the M Task (report of movement) and W Task (report of intention/urge). Prior to arrival, participants had been randomly assigned to either the M-First group or the W-First group. The M-First group completed the M Task prior to the W Task, and the W-First group completed the W Task prior to the M Task. Aside from this difference, the two groups underwent the experiment identically.

with their eyes or using some other strategy to determine its location. In addition, participants were to minimize blinking and movement during trials. Participants proceeded at their own pace through each trial and, thus, could blink and move after each individual trial. All these instructions were given in order to minimize interference with EMG recording and participants' ability to complete the tasks. Participants first completed training trials with researchers present and then completed regular trials once the researchers had left the participant room. Researchers ensured that participants had appeared to understand the tasks before leaving the room, as evidenced by following basic instructions on training trials.[10] After a task was complete, researchers returned to the room to access the next task on the computer and provide instructions, which allowed a one to two-minute break.

2.3.2. Task 1 (S Task)

The S Task consisted of five training trials and 30 regular trials. Participants viewed the clock and listened for a standard beep tone of 100 ms duration to be played through the speakers.[11] The beep was played at a random time during the 2.5–7.5 s interval after the clock hand appeared (see Jo et al., 2013). Participants then clicked the mouse any time after they heard the tone in order to stop the rotation and bring up the next screen to report where the clock hand was at the moment they first heard the beep. Researchers emphasized that the task was not about reaction time to the beep (i.e., how quickly they clicked after the beep) but about monitoring and reporting the clock hand's location in relation to the beep.[12]

2.3.3. Tasks 2 and 3 (M Task and W Task)

Participants then completed the "M Task" and "W Task," both of which consisted of five training trials and 60 regular trials. The M Task consisted solely of M reports and the W Task solely of W reports. As discussed, the M-First and W-First groups completed these two tasks in different orders (Fig. 2). In both tasks participants were asked to monitor the clock hand and make a spontaneous voluntary mouse click at the time of their choosing. They were instructed, however, per standard procedure, to wait for the clock hand to make at least one full rotation from its starting location before they clicked. Researchers emphasized to participants that the mouse clicks needed to be spontaneous and not preplanned.

Following the click, participants were to make a report. For the M Task, participants reported where the clock hand was at the moment they first started moving their hand to click the mouse. This was done in accordance with the recommendations of Pockett and Miller (2007). These authors found a significant difference in reports of movement depending on whether the beginning or the

[10] We did not "train" participants to make reports in a certain way or with certain timings, nor did we give feedback that participants had made accurate reports. We simply made sure participants were following basic instructions and were not confused.

[11] Due to equipment limitations, we used an auditory rather than tactile version of the S Task.

[12] It was an unfortunate limitation that the task required this unimportant click. We were not able to program the clock hand to automatically stop its rotation after the beep.

end of a movement was emphasized in instructions to participants. For the W Task, participants reported the moment they first had the intention/urge to move. Because it seemed reasonable to extend the findings of Pockett and Miller (2007) from movement reports to intention reports, researchers emphasized that these reports should indicate the *earliest* moment of the intention/urge to move. Crucially, when giving instructions for the second task, researchers were deliberate in not giving any indication that there would be a separate event to report in the third task. That is, researchers only gave instructions for making the voluntary movements and for the specific report type to be made.[13]

Participants then completed the third task (i.e., M Task or W Task). Researchers gave the same instructions outlined above for each task. After the third task was completed, researchers unconnected the EMG equipment and debriefed participants.

2.4. Data analysis

2.4.1. S Task

Due to extreme within-subjects variability on the task, we did not use S Task results to correct M and W reports as originally planned. We planned these corrections because we expected to obtain stable and distinct results on the S task, which would allow us to amend grand average M and W reports with confidence. However, extreme variability was present even after we used Tukey's 1.5*IQR rule to remove outliers, which resulted in the omission of between 0 and 4 trials (M = 1.0) from each participants' set of 30 trials. Overall, we omitted 22 out of 660 trials (3.3%). After outliers were removed, individual means ranged from −67 to 195 ms (M = 44, SD = 59) and individual standard deviations ranged from 38 to 216 ms (M = 86, SD = 41). The variability within participants' reports was likely due to the limited number of trials collected and the inherent imprecision of making reports using a small analog clock with a swiftly rotating clock hand (see Limitations and Recommendations for criticism of the clock). We decided it was optimal to allow this variability to remain in the reports of the other two tasks rather than to correct all reports by an average error amount with a large standard deviation. Furthermore, Dominik et al. (2017) did not collect data on the S Task, so using S Task results to correct M and W reports would have been a departure from their methodology.

2.4.2. EMG

We wished to compare results when EMG data were taken into account (EMG-corrected dataset) to when EMG data were not taken into account (EMG-uncorrected dataset). To create the EMG-corrected dataset we used EMG data to correct digital trigger timestamps, such that timestamps would indicate the objective time that movement began. We made these corrections on mouse clicks corresponding to voluntary movements on all M and W trials (but not S trials, wherein the objective time of a tone, not a click, was important). Preliminary visual inspection of EMG timings showed that correcting all trials by a standardized amount – whether across all participants or for individual participants – would be inappropriate, as discrepancies between digital triggers and EMG onset times were extremely variable. In addition, despite various attempts, we were not able to contrive a reliable software-based method of detecting threshold-crossing of EMG signals on a trial by trial basis. This was due in large part to the faint nature of what we were trying to measure (i.e., the earliest moment of movement, not the click itself). Consequently, in order to optimally determine EMG onset time we used only raw, untransformed data and manually inspected individual trials. The criterion for determining onset was a significant and prolonged departure from the preceding baseline of activity (Fig. 3). Nearly all EMG onset times were between ~70 and 200 ms prior to the triggers. Trials with EMG times outside this range were omitted from data analysis (i.e., not included in calculations for the M Task and W Task), as were trials on which it was unclear when EMG activity began (e.g., a noisy signal). In all, 565 out of 2640 total trials (21.4%) were omitted after EMG analyses, leaving 2075 trials on the EMG-corrected dataset.

2.4.3. Statistical model and normality

All other data cleaning and data analysis procedures were conducted on both datasets.[14] Following the example of Dominik et al. (2017), we removed outliers from each participant's set of 60 trials for the M and W Task separately, according to Tukey's 1.5*IQR rule. This resulted in the removal of between 0 and 7 trials per task (M = 1.82) for the EMG-corrected dataset and the removal of between 0 and 12 trials per task (M = 2.64) for the EMG-uncorrected dataset. For the EMG-corrected dataset we removed a total of 80 out of 2075 trials (3.9%), and for the EMG-uncorrected dataset we removed a total of 116 out of 2640 trials (4.4%).

We conducted a 2 × 2 Mixed Measures Analysis of Variance (ANOVA)[15] with a between-groups factor of Group and a within-groups factor of Task.[16] There were two groups, M-First (n = 10; 2 males; M_{age} = 20.3 years) and W-First (n = 12; 4 males; M_{age} = 24.42 years), and two tasks (M Task and W Task).

[13] There are discrepancies among studies concerning the term used to describe the impetus to make a voluntary movement, as well as what is reported in W trials. Researchers used both the words "urge" and "intention" - the two most commonly employed words - when verbally describing the spontaneous click to be made in both the M Task and W Task, as well as the report to be made in the W Task. For the W Task, researchers used the phrase "report where the clock hand was when you first had the intention, or urge, to click." The on-screen instructions for both tasks stated, "click when the urge arises," and the instructions for the W Task stated, "report the time when you first had the intention to click."

[14] Both datasets have the same participants (N = 22).

[15] Dominik et al. (2017) employed a Linear Mixed-Effects Model and Bayesian analysis. However, to best account for within-groups effects, we employed an ANOVA. In order to maintain a strictly Fisherian approach and confine analysis to the ANOVA, we did not employ Bayesian analysis. In place of Bayes Factors, we have reported effect sizes obtained within the ANOVA.

[16] Dominik et al. (2017) termed these factors "Order" and "Condition," respectively. We employ our own terms throughout this paper, including when we discuss the results of Dominik et al. (2017).

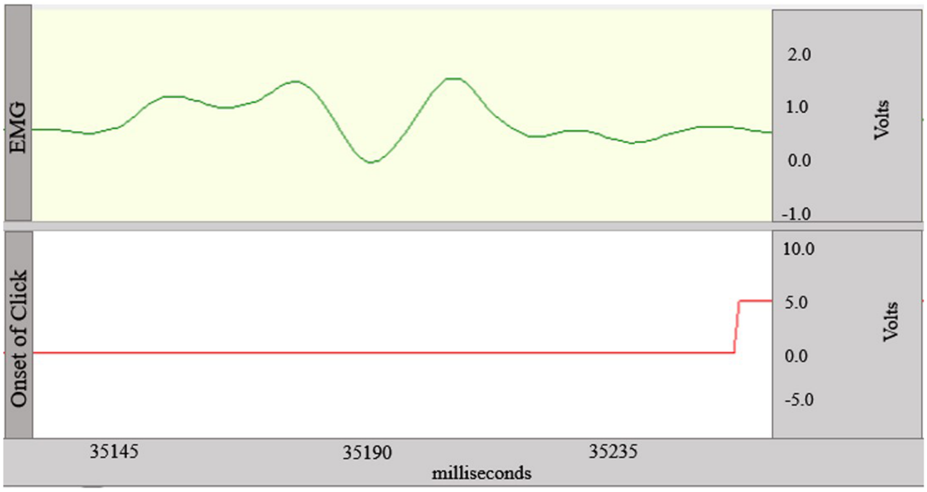

Fig. 3. EMG onset compared to digital trigger onset. On this trial, EMG activity (top pane) began (i.e., departed from baseline) about 118 ms prior to trigger onset (bottom pane).

Our method and data cleaning procedures were aimed at ensuring data met the assumption of normality. We analyzed the model residuals for normality using the Shapiro-Wilk test at $\alpha = 0.05$. For the EMG-corrected dataset, two of the four Group × Task interaction factors had normal residuals, while two did not: the M Task of the W-First group, $W(12) = 0.833$, $p = .023$ and the W Task of the M-First group, $W(10) = 0.833$, $p = .036$. A closer inspection of the residuals revealed that the former of these had a skewness of -1.807 (SE = 0.637) and kurtosis of 4.098 (SE = 1.232), while the latter had skewness of -1.446 (SE = 0.687) and kurtosis of 1.604 (SE = 1.334). Thus, in both cases there appears to be slight negative skew and slight positive kurtosis (i.e., leptokurtosis). The assumption of normality was met for all Group × Task interaction factors for the EMG-uncorrected dataset.

Using Levene's test at an alpha level of 0.05, we assessed homogeneity of variance across groups on both tasks. The error variance was found to be equal across groups for the M Task on both datasets but unequal across groups for the W Task on the EMG-corrected dataset, $F(1, 20) = 5.855$, $p = .025$ as well as the EMG-uncorrected dataset, $F(1, 20) = 6.294$, $p = .021$.

3. Results

3.1. EMG-corrected dataset

The model showed no main effect of Group, $F(1, 20) = 0.899$, $p = .354$, $\eta_p^2 = 0.043$, nor a main effect of Task, $F(1, 20) = 1.703$, $p = .207$, $\eta_p^2 = 0.078$.[17] There was a significant interaction of Group and Task, $F(1, 20) = 6.607$, $p = .018$, $\eta_p^2 = 0.248$. Therefore, group membership, which determined the order of M Task and W Task completion, influenced the tasks differently. These findings were in line with those of Dominik et al. (2017).

Pairwise comparisons were conducted within the ANOVA via analyses of simple effects ($\alpha = 0.05$, Bonferroni corrections for multiple comparisons; see Fig. 4). All possible between-groups and within-groups pairwise comparisons were conducted. Means were significantly different on only one comparison: in the M-First group, W reports ($M = -34$)[18] were significantly earlier than M reports ($M = 121$), $F(1, 20) = 6.883$, $p = .016$, $\eta_p^2 = 0.256$. This finding provided support for our second hypothesis. Our second hypothesis was provided further "support" by the finding that there was *not* a significant difference between M reports ($M = 61$) and W reports ($M = 111$) in the W-First group, $F(1, 20) = 0.881$, $p = .359$, $\eta_p^2 = 0.042$. Note that because we anticipated a null effect on this comparison, a non-significant difference between groups cannot, strictly speaking, be said to support the second hypothesis (i.e., this result is not conclusive evidence that there is no difference in performance on the two tasks). The relatively small difference in means (~50 ms) and small effect size (4%) further suggest, however, that there is likely no difference on this comparison. The first hypothesis was likewise provided "support" by the lack of a significant difference between groups on the second task completed (i.e., the task following the S Task): the M reports of the M-First group ($M = 121$) and W reports of the W-First group ($M = 111$) did not differ, $F(1, 20) = 0.077$, $p = .785$, $\eta_p^2 = 0.004$. Again, this result is not conclusive evidence that there is no difference between groups on this comparison. In this case, the extremely small difference in means (~10 ms) and miniscule effect size (0.4%) further suggest no difference between groups.

On the W Task, the M-First group ($M = -34$) and the W-First group ($M = 111$) showed a mean difference of nearly 150 ms, but this difference just failed to reach significance, $F(1, 20) = 3.906$, $p = .062$, $\eta_p^2 = 0.163$. Therefore, our third hypothesis was not

[17] The Group main effect is the comparison between groups using the combined results of the two tasks. The Task main effect is the comparison between tasks using the combined results of the two groups.

[18] All values reported hereafter are in milliseconds (ms) and are relative to the time of movement. Thus, positive numbers reflect "late" reports (made after movement), while negative numbers reflect "early" reports (made before movement).

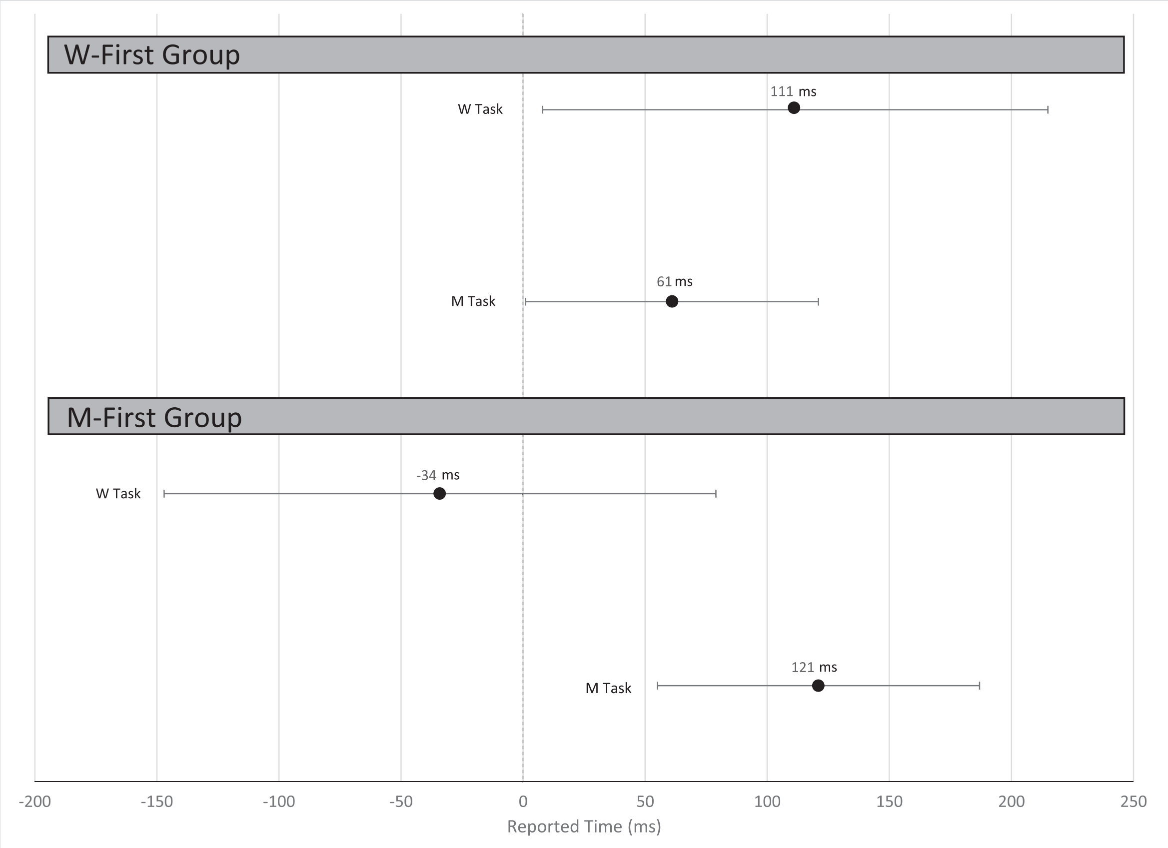

Fig. 4. Reported times for the EMG-corrected dataset. The dots signify the estimated marginal means (in milliseconds) of the Group × Task factors. The bars reflect 95% confidence intervals.

supported. This particular finding of non-significance was a departure from the results of Dominik et al. (2017). Their results showed that W reports in the M-First group were significantly earlier than *all* other reports; W reports (M = −286) were significantly earlier than M reports in the same group (M = −8), as well as the W-First group's M reports (M = −52) and W reports (M = −11). However, as noted, only one pairwise comparison was significant in the current study, between M and W reports in the M-First group.

3.2. EMG-uncorrected dataset

All results on this dataset mirrored the EMG-corrected dataset in terms of significance and the support and/or "support" provided for each hypothesis. Thus, the findings on this dataset corroborated those of Dominik et al. (2017) regarding no significant main effect of Group, $F(1, 20) = 1.088$, $p = .309$, $\eta_p^2 = 0.052$ or Task, $F(1, 20) = 1.860$, $p = .188$, $\eta_p^2 = 0.085$ but a significant interaction effect, $F(1, 20) = 6.990$, $p = .016$, $\eta_p^2 = 0.259$.

All pairwise comparisons showed the same significance as in the EMG-corrected dataset, but, of course, means differed (Fig. 5). In the M-First group, W reports (M = −174) were significantly earlier than M reports (M = −13), $F(1, 20) = 7.362$, $p = .013$, $\eta_p^2 = 0.269$. In the W-First group, there was no difference between M reports (M = −70) and W reports (M = −20), $F(1, 20) = 0.901$, $p = .354$, $\eta_p^2 = 0.043$. Regarding the second task completed, there was no difference in M reports of the M-First group (M = −13) and W reports of the W-First group (M = −20), $F(1, 20) = 0.031$, $p = .862$, $\eta_p^2 = 0.002$. There was a mean difference of over 150 ms on W reports between the M-First group (M = −174) and the W-First group (M = −20), but this difference was on the threshold for non-significance, $F(1, 20) = 4.358$, $p = .050$, $\eta_p^2 = 0.179$.

4. Discussion

4.1. Replication of Dominik et al. (2017)

The current study attempted to replicate the results of Dominik et al. (2017), which suggested that the intention reports developed by Libet et al. (1983) do not reflect an actual experience of intention but, rather, are invalid inferences made as a result of prior training and experience with movement reports. On the whole, our results corroborated the core findings of Dominik et al. Results

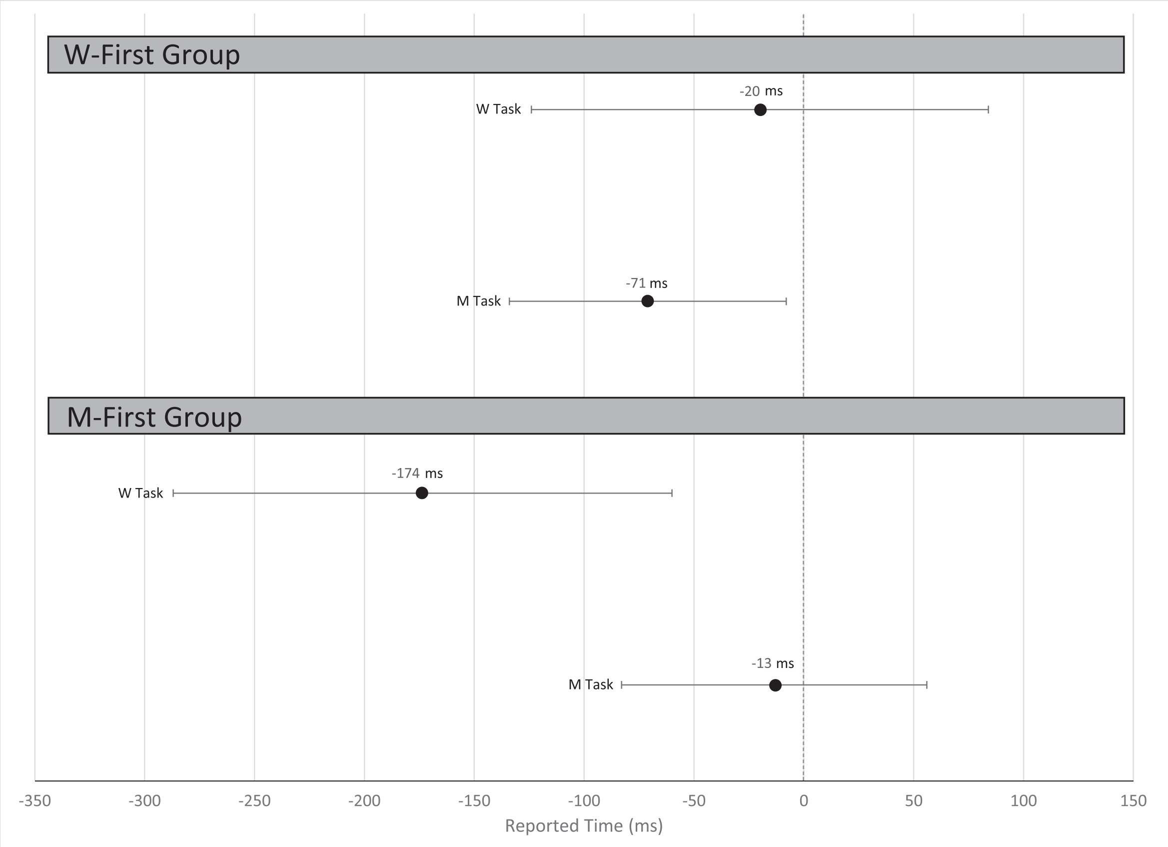

Fig. 5. Reported times for the EMG-uncorrected dataset. The dots signify the estimated marginal means (in milliseconds) of the Group × Task factors. The bars reflect 95% confidence intervals.

from both our datasets agree with Dominik et al. regarding the non-significance of the main effects of Group and Task and the significance of their interaction. We thus replicated the order effect in which order of movement (M) Task and intention (W) Task completion (determined by group) influenced tasks differently. Furthermore, we replicated the non-significant between-groups difference between W and M reports when participants had no prior experience with or knowledge of the two types of report, as well as the significant difference between M and W reports in the M-First group (i.e., W was reported significantly earlier). These two comparisons were the most crucial evidence indicating that participants make intention reports earlier in time than movement reports *only* when they have prior experience with movement reports to "anchor" intention reports. In addition, we replicated the finding that participants without such experience (i.e., the W-First group) made indistinguishable intention and movement reports, which lent further support to the idea that experience with movement reports causes anchoring. However, as discussed, the non-significant differences crucial to our study do not conclusively rule out the possibility of a genuine difference on each comparison. Additionally, note that on the second task, both M and W reports were around 115 ms for the EMG-corrected dataset but around −15 ms for the EMG-uncorrected dataset. Dominik et al. (2017) found grand average means around −10 ms on this comparison. Thus, these means are similar between studies when considering the EMG-uncorrected dataset, but there is a discrepancy of over 100 ms when considering the EMG-corrected dataset. This is likely because Dominik et al. reported data uncorrected by EMG.

Our results depart from those of Dominik et al. (2017) regarding the comparison of W reports of the M-First group to the other reports. Whereas Dominik et al. (2017, pp. 259-260) found that M-First W reports were significantly earlier than all other reports, we found that they differed only from M reports in the same group, although the ∼150 ms difference between them and W reports of the W-First group approached significance in both datasets. Dominik et al. pointed to the significant discrepancy in W reports between groups (∼275 ms) as further evidence that previous experience with M reports in the M-First group changes the timing of W reports. While we did not corroborate this finding, the difference was clearly in the correct direction and might have reached significance with additional participants and/or trials.

Considered as a whole, our results replicate the core findings of Dominik et al. (2017), to which the aforementioned discrepancies are peripheral. The replicated findings are sufficient to confirm the original doubts raised by Dominik et al. about the validity of intention reports in the Libet experiment. The Libet experiment's methodology produces reports of intention, but these are apparently based on inferences rather than experiences of intention. It is worth noting that we achieved a replication despite including the S Task. Dominik et al. (2017, p. 262) recommended that a separate group be given a "dummy task" in future studies to determine whether W reports are uniquely influenced by previous experience making M reports or might be influenced by *any* task that employs clock-monitoring. Instead of using separate groups, we had both groups first complete the S Task. Since previous experience with this "dummy task" (i.e., the experience of being trained on and completing the task) did not prevent us from replicating the core findings of Dominik et al., particularly the timing of W reports in the W-First group, we can confidently affirm the conclusion that experience with M reports uniquely influences (i.e., anchors) W reports.

4.2. Interpretations

We have corroborated the core findings of Dominik et al. (2017) and accepted their general conclusion that intention reports in the Libet experiment are invalid because they are based on inference via the anchoring effect. However, we do not necessarily concur with their interpretation that intention is not experienced in the movements required by the experiment. Although it may seem that intention is altogether absent from the spontaneous voluntary movements in the Libet experiment (i.e., not experienced), it is possible that intention is indeed experienced in these movements. In this case, questions arise about "when" intention is actually occurring and the general possibility of using reports to measure intention. We discuss each interpretation in turn. Note that neither interpretation challenges the invalidation of Libet's intention reports. Each simply seeks to account for the phenomenology of intention in the Libet experiment in light of the anchoring effect. The following discussion may provide some contribution to discussions of free will, the self, or related topics, but treatment of such topics is outside the scope of this paper.

4.2.1. Interpretation 1: Intention is not experienced

On this view, individuals do not make a natural distinction between moving and intending to move, at least in the context of the Libet experiment, because they do not experience "intention" when carrying out the required movements. Dominik et al. (2017, pp. 259-261) argued that there is only a single "introspective phenomenon" present during the M and W Task: the experience of movement (see Introduction). This provides a plausible explanation as to why participants in the W-First group, when allowed to make intention reports reflecting their natural phenomenology, yielded reports indistinguishable from *movement* reports, both of which were near the objective time of movement. [19]Compare this to the results of Libet et al. (1983), in which intention was reported more than 100 ms earlier than movement, and only movement reports occurred near the objective time of movement. If "natural" intention reports appear to be indistinguishable from movement reports, the most parsimonious explanation might be that there never was any "intention" to report in the first place. Moreover, the lack of a genuine experience of intention might be a necessary condition for the anchoring effect.

Why might intention be absent from the movements carried out in the Libet experiment? Many questions have been raised about the *ecological validity* of the "intentional movements" asked of participants, leading to skepticism about accepting the findings of the Libet experiment and/or extrapolating them to everyday life (e.g., Banks & Pockett, 2007; Mele, 2013; Pitman, 2013; Navon, 2014; Kihlstrom, 2017). It seems plausible that the movements are too simple or artificial to actually involve "intention" as commonly experienced. The movements are voluntarily initiated, but they seem to be so stripped of purpose and context that they involve no deliberation or decision-making (i.e., intention). Participants are told to simply move spontaneously without preplanning, which seems to preclude the possibility of any "intentional" deliberation. It is worth asking whether "intention" can be phenomenally present in such movements. It seems paradoxical to "intend" a movement while simultaneously acting "spontaneously" without preplanning. Did the Libet experiment try to measure intention despite instructing individuals to move without intention? This may accord with the skepticism voiced by Nachev and Hacker (2014) that the Libet experiment makes the implicit yet unproven assumption that a conscious intention *always* precedes a spontaneous voluntary movement.

It should be noted that those variants of the Libet experiment which have attempted to make the spontaneous movement more meaningful and deliberate, such as by allowing participants a choice between responding with the left or right hand, have achieved more or less the same overall results as the Libet experiment (e.g., Haggard & Eimer, 1999). However, as discussed by Pitman (2013) the concerns regarding ecological validity extend to these variants of the Libet experiment as well. No matter how much one attempts to bolster the spontaneous voluntary movements, they remain "meaningless and decontextualized bodily movements" which are not what actions anyone would "normally engage in, nor…typically 'feel the urge' to do" (Pitman, 2013, p. 86). In short, the very nature of the Libet experiment, in all its variants, is to severely constrain movements. Because movements must occur spontaneously without preplanning, they may exclude the intention they are supposed to involve.

Papanicolaou (2017, p. 314) distinguishes between *acts* and *mere movements*. For Papanicolaou, acts are intentional in that they are carried out for some meaningful goal or purpose, while mere movements are not. What is asked of participants in the Libet experiment, according to Papanicolaou, is a mere movement with no consequence or significance other than in the minimal sense of continuing a research study. The mere movements of the Libet experiment, in a decontextualized setting, can give the false appearance of being meaningful acts because some acts involve similar movements. After all, to act one must usually move in some way,

[19] This refers to the results of Dominik et al. (2017) and our EMG-uncorrected dataset, since both were around 10-15 ms prior to movement. For the EMG-corrected dataset, these reports were around 115 ms post-movement, so although both datasets agreed in terms of non-significance between groups, the reports were not near the objective time of movement in the EMG-corrected dataset. In addition, M reports in the W-First group were also post-movement (61 ms). Note that it is likely that this dataset captured movement reports more accurately, so it may be that participants naturally make late reports of movement. Therefore, the argument of Dominik et al. (2017) - that on the second task both groups are reporting movement - is not necessarily affected by these findings.

and some acts require simple hand movements. Papanicolaou (2017, p. 314) gives the example of pressing a button to vote or buy stocks. In these cases, moving to press a button is accompanied by a host of mental processes, such as those for planning, deliberation, judgment, attention, impulse control, and the like. In the Libet experiment, there is a movement to press a button, but it is not supposed to involve these mental processes. One is supposed to simply move spontaneously, and one's attention is to be focused on following the rules for using the clock to make reports. Simply put, the Libet experiment is deliberately designed to produce simple, restricted mere movements rather than acts, and these movements may be devoid of intention.

The movements in the Libet experiment, whether they be mouse clicks, wrist flexes, or some variation thereof, are certainly voluntary, but perhaps only in the most minimal sense. The term "voluntary" seems to entail intention, but the results of the current study suggest that some voluntary movements may not be "intentional" in the normal sense. Many authors have pointed out that in the Libet experiment the only "decision" to be made is about *when* to act, which is supposed to somehow occur without prior planning or deliberation (e.g., Haggard, 2008; Schmidt, Jo, Wittmann, & Hinterberger, 2016; Papanicolaou, 2017). The decision *to* act is already made when the participant begins individual trials; by understanding instructions and agreeing to complete blocks of trials, the participant has already decided *that* and *how* he/she will move, and the only question is *when* to move. At best, the M Task and W Task involve when-decisions (Haggard, 2008; Schmidt et al., 2016). These decisions seem very different from everyday decisions associated with voluntary acts of will. Intention may simply be phenomenally absent in when-decisions, or, at least, in those made in a decontextualized experimental environment.

Furthermore, many authors have argued that there is not a positive phenomenology associated with agency (e.g., Paglieri, 2013; Chadha, 2016). On this view, there is no "agentive" feeling that accompanies voluntary (i.e., intentional) movements or acts; one simply moves or acts voluntarily. Chadha (2016, p. 203) noted that while there is a phenomenological feeling of the *loss* of authorship of one's actions in cases of abnormal agency, such as "alien control" in schizophrenia, this does not entail that there was a primal feeling of normal agency that was lost or replaced (for related examples of non-pathological loss of agency, see Farrer et al., 2003; David et al., 2007). Simply put, one only experiences a feeling associated with agency if something goes awry during the process of agency. If all goes well, there is no salient feeling that one is the author of one's actions; one simply *is* the author of one's actions. Per this general account of the phenomenology of agency, movements in the Libet experiment may be voluntary yet devoid of reportable intention because they lack positive phenomenological content.

4.2.2. Interpretation 2: Intention is experienced

On this view, intention is experienced, in some manner, in the voluntary movements carried out in the Libet experiment. This interpretation accords with the general claim that there *is* a positive phenomenology of agency. Proponents of this claim argue that there is always some feeling(s) associated with voluntary movement, though they do not agree on the qualities and degree of salience of such feeling(s) (for discussion see Gallagher, 2012; Chadha, 2016). There are two versions of this interpretation. The first version interprets the results of the current study and those of Dominik et al. (2017) as indicating that intention occurs at more or less the same time as movement. The second version holds that intention is experienced but cannot be accurately measured due to misreporting and/or other issues.

First, it may be that intention is experienced in the required voluntary movements and that the methodology developed by Dominik et al. (2017) allowed intention to be accurately measured when participants did not have prior experience with movement reports (i.e., on the second task completed in the current study).[20] That is, it may be that the intention reports in the W-First group of the current study and Dominik et al. (2017) accurately reflect experiences of intention. It seems far from implausible that in spontaneous voluntary movements, intention could more or less coincide with movement. Without being unduly influenced to make inferences about intention, individuals may simply be reporting intention when they experience it. Some participants gave us feedback strongly suggesting that they may have experienced intention in this way. Many reported that they did not experience a gap between intention and movement, and some were unsure if there was a gap (though some reported they did not always feel an intention to move before they moved). As for the intention reports of the M-First group, on this interpretation the anchoring effect must have had enough cognitive influence to override the natural experience of intention. That is, anchoring may be possible even if intention is experienced. This account is likely more plausible if the phenomenology associated with agency is "thin or minimal" as some contend (Tsakiris, Schütz-Bosbach, & Gallagher, 2007, p. 659). It should be noted that this version of the interpretation, while still invalidating the intention reports in the Libet experiment, could potentially strengthen the core findings of the Libet experiment, since it might place intention even later in time past the RP onset. If intention is indeed occurring closer to the time of movement, then perhaps the gap between the RP and intention is even larger than it originally appeared.

Second, it is possible that intention is present in the movements but is affected by any and all attempts to measure it (i.e., observe and report it). Consequently, it may be impossible to capture accurate reports of intention. Furthermore, even if it were possible to capture accurate reports, it might be impossible to verify that a study has done so. In short, this interpretation of results grants the presence of intention in the tasks but questions the validity of all intention reports, including those in the current study. On this view, not only are W reports of the M-First group invalid due to the anchoring effect, but the W reports of the W-First group are also suspect. Simply put, their accuracy is indeterminable.

For one, intention, like other psychological phenomena, is likely to be affected by the attention focused upon it. An observer effect

[20] Note that there is a discrepancy in these reports between the results of Dominik et al. (2017) and the current study. These reports were at −287 ms in Dominik et al. (2017). In the current study, these were at −174 ms in the EMG-uncorrected dataset and −34 ms in the EMG-corrected dataset.

appears to be at play here, in which attempts to introspectively examine an ongoing mental process inevitably change the process.[21] To use Heideggerian concepts, intention may be inextricable with one's purposive, practical *engagement* in movement, and it may be impossible to accurately abstract about the *occurrent* features of intentional movement (Käufer & Chemero, 2015, pp. 50-76). In other words, it is one thing to engage in an intentional movement but another to try to introspect during the ongoing engaged process and then attempt to divulge information from abstractions. Similar to how the very act of measuring one's blood pressure can often change blood pressure and yield inaccurate readings (Dent, 2013), one may not be able to attend to intention (i.e., observe and/or report on it) without changing its phenomenology, such as the experience itself and the perception of its "onset." As Wasserman (1985) argued, the act of reporting a conscious intention is itself an additional voluntary act.

The insertion of additional variables and acts into one's natural phenomenology likely adds up to a considerable cognitive load for participants during tasks. The tasks, despite appearing to consist of only a few simple components, are in fact attentionally demanding (Fig. 6). Participants are asked to comply with a litany of rules and requests, some overarching for the entire study and some particular to individual tasks. All of these insertions may interfere with one's natural intentional engagement, one's experience of intention, and/or one's ability to report on intention.

Unfortunately, if the onset of intention is indeed misreported due to such a process of interference with one's natural phenomenology, it may be impossible to know in which direction this is occurring. Papanicolaou (2017) noted how the presence of multiple demands for attention during tasks can lead to the well-established *effect of prior entry*, in which multiple simultaneous events cannot occur all at once. He noted the demand for at least three simultaneous acts during Libet-style tasks: making a movement, observing the clock, and remembering the location of the clock hand at the time of an event. He argued that the presence of the effect of prior entry means that only one of these events can occur at a time. It is unclear – and perhaps impossible to determine – which two of the three events are delayed in time. Thus, even if we forego other reservations about individuals' ability to accurately perceive and report on his/her experiences, this effect indicates that only one event can be perceived at a time, leading to an unknown amount of error in perception and/or reporting for all other simultaneous events. Since there are many additional ongoing demands like those previously discussed, this effect may be significant.

Papanicolaou (2017) also discussed the unclear distinction between activity and passivity in the Libet experiment. He noted that it is often assumed that participants are passive observers of their mental states and, thus, can make objective reports about them. It is as if participants can objectively observe their mental states without interference or prejudice in the same way that, for example, astronomers are thought to observe the night sky. Papanicolaou (2017, p. 314) emphasized that participants are active creators of their mental states, noting that they "make choices: choices do not 'happen' to them." Though the assumption of passivity is obviously erroneous, Papanicolaou argued that it has been taken for granted since the original study by Libet et al. (1983). By undercutting this assumption, Papanicolaou undermined the "objective" status of the intention reports made in any form of Libet experiment. The fact that observing and reporting on intention is active rather than passive further highlights the aforementioned observer effect, in which the attempt to observe and report on a subjective phenomenon changes the phenomenon and/or the ability to make accurate reports about it.

As discussed, there is a lack of consistency in phrases and terms for how intention reports and "intentions" in the context of voluntary movements are framed to participants. Because of this, researchers in the current study deliberately used the two most common words, urge and intention, when referring to making spontaneous movements and to making reports on the "W Task."[22] It seems reasonable that considerable individual differences exist in subjective understandings of these complex words, especially in the context of the Libet experiment. For example, does urge imply a bodily component that is not inherent in the concept of intention, and does urge imply more passivity than intention? Feedback from participants indicated a wide range of opinions as to what they thought the words meant, as well as what was being reported in intention reports. Researchers may wish to clearly define and explain any terms/concepts used in an experiment, but this leads to a further issue, related to the original question raised by Dominik et al. (2017) and addressed in this study: does providing *a priori* distinctions, like definitions, induce demand characteristics and/or change the phenomena in question? This is an especially important consideration given that individual differences might also exist in the phenomenology of carrying out intentional acts and/or introspecting on them (Nahmias, Morris, Nadelhoffer, & Turner, 2004).

Richards (2010, pp. 8-10) has discussed how applying language to psychological phenomena is a highly reflexive enterprise. Simply put, for language about private psychological states to be meaningful, it must be based on "external world categories" (e.g., metaphors), yet how one labels, discusses, and thinks about psychological phenomena is likely to affect the phenomena (Richards, 2010, pp. 160-161). In other words, as the language describing our psychology changes, so does our psychology. Richards' argument primarily focuses on general trends in Psychology and society, such as the introduction of new concepts (e.g., IQ) or gradual adoption of new terms (e.g., stress instead of fretting), but his points are extendable to any intersubjective situation in which one is referring to psychological phenomena. In our case, even if researchers try to remove individual differences in understandings of terms/concepts, this may change the phenomenon being measured and/or induce confusion, thereby invalidating measurements. This appears to be another observer effect; the more a natural subjective experience or understanding is tampered with, the more it is likely to be unduly biased or changed. Furthermore, a more fundamental issue looms; how could researchers ever know participants were conforming to what they have in mind for "urge" or whatever term-construct is chosen?

The aforementioned issues apply to *all* attempts to garner reports about intention, including the methodology of Dominik et al.

[21] Miller, Shepherdson, and Trevena (2011) noted the possibility of observer effects in the Libet experiment. These authors discussed how clock monitoring is necessary for reports yet may change, or even cause, the RP.

[22] Recall that "W" stands for wanting, an additional term with its own connotations which was introduced by Libet et al. (1983).

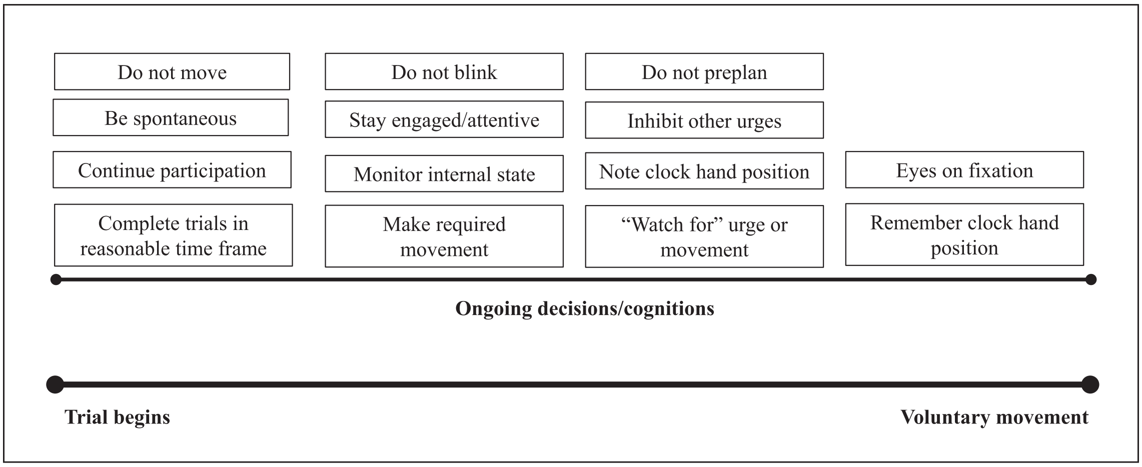

Fig. 6. Visual representation of cognitive load during tasks. Participants are asked to engage in several attentionally demanding acts and cognitions throughout task completion, which likely interferes with the act of reporting intentions.

(2017); *any* measure that affects a natural intentional process seems prone to inaccuracies in reports of intention. Thus, intention may be present but necessarily unamenable to measurement via reports, no matter the methodological design (e.g., the W reports of the W-First group). Furthermore, since intention is probably best thought of as a dynamic process spread out in time rather than a discrete, phenomenologically clear event (Banks & Pockett, 2007; Verbaarschot, Haselager, & Farquhar, 2016), the question of "when" intention begins may not be completely coherent. Pitman (2013, p. 75), drawing on the work of Strawson (2004), argued that we do not "have" mental states like intentions, nor are such mental states "in us." Instead, "we are *in* these states" in the sense of occupying them as agents carrying out an ongoing action (Pitman, 2013, p. 75). On this view, it may be misleading to treat such mental states as discrete and isolable. When we engage intentionally in a "temporally extended intentional framework" our mental states have boundaries which are "fuzzy, continuous, and overlapping with a range of other mental and bodily processes that evolve over time" such that the concept of initial awareness of the urge to move may be incoherent and meaningless (Pitman, 2013, p. 76; p. 86). For Pitman, this applies to everyday intentional action as well as the spontaneous movements carried out in the Libet experiment and its variants. The attempt to measure intention as a clear, distinct event may yield empirical results, but those results may inevitably be inaccurate and misleading. Different methodologies may necessarily produce different empirical results because each inserts a distinct set of variables into participants' cognitive load, thereby uniquely influencing the experience, and/or reporting, of intention. [23]If our experiences associated with agency are indeed "thin or minimal" (Tsakiris et al., 2007, p. 659), these issues of interference are likely more substantial. The feeling associated with intending to move is probably more susceptible to undue influence if it is naturally faint, vague, and fleeting, rather than distinct and robust.

The issue of *inferential distance* is lurking here. These measurement problems are a fundamental problem for the Libet experiment because it relies on reports of intention rather than directly measuring intention. It is a rather obvious point that the Libet experiment indirectly attempts to measure intention, and that the goal of Dominik et al. (2017) and the current study was to determine the validity of this attempt, but it is a point worth emphasizing. If there were some universally accepted method of directly measuring intention, then the Libet experiment would not have to rely on reports of intention. In the absence of an unequivocally established index of intention (if such an index is possible), reports of intention, and the inferential distance between report and phenomena, are necessary. Furthermore, the previously discussed issues regarding task interference with one's natural phenomenology reflect a secondary process of inferential distance. It is one thing to carry out an intentional action but quite another to introspect and report on that action, so there is inferential distance between the experience of intention and the report thereof. Therefore, there are two types of inferential distance between the phenomenon targeted for measurement (i.e., intention) and the data yielded about that phenomenon (i.e., intention reports). The reports are not only indirect measurements of intention but also imperfect representations of the experience of intention. It would be a mistake, however, to claim that one's inability to accurately report intention indicates the absence of intention. Surely one can have an experience without being able to accurately report on it.

Finally, worth mention is a line of research and theory suggesting that the "sense of agency," or the feeling that one intended and controlled an action, may be constructed/inferred after action (e.g., Frith, Blakemore, & Wolpert, 2000; Farrer & Frith, 2002; Bays, Wolpert, & Flanagan, 2005; also see Introduction). This account of the *origin* of agency, typically termed the "comparator model," contends that agency is "inferred" and, thus, appears to have a close connection to the current study. Although *prima facie* it may seem that our findings regarding "inferences" fall under the model and/or that the model might clear up ambiguities in interpreting results, this is not the case. Per the comparator model, the sense of agency arises only when there is a sufficient match between the anticipated effects of an action and the actual effects of the action in the external world, which includes one's body. One cannot form a sense of agency – of having been the intentional author of an action – until *after* one has already carried out an action and processed

[23] Of course, intention may be altogether absent as well, and different methods may simply produce different artificially induced reports.

its effects (e.g., visual, proprioceptive, and/or motor signals), which allows a *post hoc* reconstruction of agency (Chambon, Sidarus, & Haggard, 2014). This account of agency deals with *how* a sense of agency is generated. However, the aforementioned issues regarding the phenomenology and measurement of intention apply regardless of how the feeling of intention arises in the first place.

First, if intention is experienced in the Libet experiment movements, this is so regardless of how originally "arose." Second, given the demonstrated anchoring effect, it seems that even if the comparator model were true, the experience and/or reporting of intention is easily manipulable. When the M-First and W-First groups made intention reports the only difference between groups was the prior experience with movement reports in the M-First group. There was no difference between groups in manner of reporting or movement, nor in post-movement effects in the external world. Thus, there were no external differences (e.g., in proprioceptive signals) to create a difference between M-First and W-First intention reports. The only mechanism to produce this difference was the discrepancy in prior experience, which was represented internally. In short, even if intention is a *post hoc* reconstruction, it is subject to additional forces after it arises phenomenally. Therefore, the comparator model does not seem of much service in helping us resolve the issues surrounding the phenomenology and measurement of intention. Moreover, though the current study and the comparator model both refer to inferences, these are clearly not inferences of the same type. In the current study "inferences" from anchoring are purely cognitive, though not necessarily conscious, and they pertain to intention reports. In the comparator model "inferences" may involve cognitive components but also must involve external feedback (e.g., visual perception), and they pertain to the construction of intention itself. Note also that the comparator model has been challenged. Chambon et al. (2014), for example, provided evidence that mental processes related to the fluency of action selection, which occur well *before* the effects of the action, play a role in generating a sense of agency. It appears, then, that the comparator model, instead of aiding us in interpretation, further muddies the waters.

4.3. EMG variability

Many aspects of the EMG results were surprising. We expected to find much less within- and between-subjects variability in EMG onset times relative to digital triggers. We assumed that participants would make a simple mouse click in more or less the same way on each trial. Data analysis quickly revealed more than expected variability in the EMG, and it was necessary to manually determine movement onset trial by trial. This seemingly mundane finding regarding calculations for movement onset may be extremely important for the Libet experiment. The norm appears to be the exclusion of detailed information regarding how the moment of movement initiation was measured, though Jo et al. (2013), for example, noted that they placed an accelerometer on the mouse itself to determine the onset of the button press. Accelerometers may be an alternative to EMG, though it would seem they capture the earliest moment a button begins moving rather than the earliest moment of bodily movement.

One is left to wonder, in the absence of detailed reporting, how often and how precisely the moment of movement initiation has been measured in previous research. If researchers have not adequately accounted for the variability in movement (i.e., by measuring the *earliest* moment of movement and ensuring individual trials are appropriately corrected) then this is a random source of error on all M and W reports and will also cause the RP to be improperly time-locked. If the objective time of movement onset is not properly calculated, each individual trial will be shifted backward or forward in time by the error. Our results recommend further testing of EMG within Libet-style experiments. At the least, our results recommend the use of EMG or some measure to capture the earliest moment of movement, as well as improved reporting of such measures.

5. Limitations and recommendations

The fact that some data violated the ANOVA assumptions of normality and homogeneity of variance is a limitation for our study. Dominik et al. (2017, p. 259) described the use of Tukey's 1.5*IQR rule to remove outliers and stated, "without the outlier removal the model residuals do not meet the assumption of the normal distribution," but otherwise did not discuss normality or homogeneity of variance. Thus, it is unclear whether Dominik and colleagues tested for normality and homogeneity of variance or assumed them based on outlier removal. The findings and conclusions of both studies should be taken with a modicum of caution until these issues can be resolved. Future studies should first and foremost increase sample size. This was one of the original recommendations of Dominik et al. which we tried to accommodate, but various issues led to less data collection than expected and the exclusion of several participants' data.

A variety of untreated factors may have contributed some variability to results. For example, we collected demographic data but did not conduct analyses involving covariates. In addition, we did not collect data on other variables which might have impacted task performance, such as working memory capacity, familiarity with computers, and frequency of using a mouse. We did not counterbalance variables such as mouse response mapping and time of day of sessions. In addition, we did not collect self-report data on preplanning during tasks, which we might have used to discard data from participants who made planned instead of spontaneous movements.

Variability is also likely present in the anchoring effect and in the effects of any "training" administered to participants. Our inclusion of training on the dummy task (S Task) may be met with the objection that results would have differed without such training. Training on this task, however, did not overlap with the essential, distinct components of the M and W Tasks (i.e., spontaneous clicks and M and W reports). The overlap pertained to instructions for using the clock and user interface. Nonetheless, a future study might devise an alternate experimental design to assess the effects of this and/or other types of training. Such a study might assess the variability of the anchoring effect under various training conditions and/or task orders.

Our study used visual stimuli from one experiment (Jo et al., 2013) but followed the methodology of another (Dominik et al., 2017). It seems plausible that small differences in stimuli could impact findings. For example, the clock used in the current study uses a rotating clock hand, while Dominik et al. used a rotating dot. The nature of the clock face also differed in many ways (e.g., color, fixation stimulus, number of intervals), though the basic setup (e.g., rotating clockwise, revolution period) was mostly identical. There is a possibility that, in the absence of identical stimuli, at least some of the variance in results may be due to differences in stimuli. Additionally, the current study used mouse clicks, while Dominik et al. used button presses and Libet et al. (1983) used wrist movements. Though all these movements appear to be highly similar, one cannot rule out the possibility that some of the differences between studies are attributable to differences in type of movement.

In addition, participant reports may appear more precise than they actually are. Each clock tick spanned approximately 16.667 ms, inside a clock face only 3.5 cm in diameter. Participants were supposed to look at the center of the clock rather than the clock hand or face. It seems unlikely that such a setup would allow participants to note and report clock hand location with accuracy on the scale of tens of milliseconds. The purpose of the S Task was to try to account for general bias in using the clock, but there was extreme variability in S Task results. This may reflect, at least in part, a human subject limitation in using a Libet-style clock. It may be that the maximum attainable temporal resolution with the clock is on the order of 50 ms or more, as reflected by S Task results. This is an important consideration for any version of the Libet experiment, since effects are on the order of hundreds of milliseconds or less.

6. Conclusions

Overall, our results corroborated the core findings of Dominik et al. (2017), and we concur with their general conclusion: intention reports in the Libet experiment are inferred as a result of prior experience with movement reports and are, therefore, invalid. The crux of the Libet experiment is the temporal relationship of the RP and intention, and if its intention reports are invalidated, then the findings regarding this relationship are also invalidated. It might be possible, however, to reaffirm the original results of Libet et al. (1983) if one interprets our results as indicating that intention occurs around the time of movement and, thus, is still significantly preceded by the RP. Of course, this approach would have to rule out the other, perhaps more plausible explanations of results holding that intention is absent from the movements or is unamenable to accurate measurement. The “Dominik experiment” is of fundamental importance for the future of the Libet experiment, and more replications and analyses of its implications are needed. A future replication might maintain the current study’s methodological additions while attempting to remedy its shortfalls. However, given this study’s contribution to the ever-growing body of literature which casts further doubt on the Libet experiment’s validity and usefulness, it is worth asking whether the time has come to phase out this research paradigm.

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Declaration of Competing Interest

None.

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